



Concrete Experience



U.S. ARMY





United States Army Cyber Center of Excellence

Electromagnetic Warfare Training Management

112-17ABOLC

OVERALL CLASSIFICATION:

UNCLASSIFIED



Learning Step Activities



U.S. ARMY

- LSA 1 - Identify Mission Essential Task List (METL) of the EW PLT
- LSA 2 - Define Required Individual and Collective Tasks within the EW PLT METL
- LSA 3 - Apply the 8-step training model to EW PLT METL
- LSA 4 - Apply Troop Leading Procedures (TLP) to the EW PLT



Administrative Information



U.S. ARMY

Safety Requirements	NONE
---------------------	------

Risk Assessment Level	LOW
-----------------------	-----

Environmental Considerations	None
------------------------------	------

Evaluation	???
------------	-----



References

U.S. ARMY

- FM 3-12. *Cyberspace and Electromagnetic Warfare Operations*. 17 Sep 2025.
- FM 7-0. *Training Management*. 14 Jun 2021.
- ADP 5-0. *The Operations Process*. 31 Jul 2019.
- ATP 3-12.3, C1. *Electromagnetic Warfare Techniques*. 18 Sep 2024.
- ATP 3-12.4. *Electromagnetic Warfare Platoon*. 09 Jan 2023.
- *Electromagnetic Warfare Training Strategy (EWTS)* v7. 01 Mar 2021.
- *TC 3-12.1.2.2. *The United States Army Cyber and Electromagnetic Warfare Training Strategy (CEWTS)*. 01 Nov 2024.



U.S. ARMY



Identify Mission Essential Task List (METL) of the EW PLT



Understanding Mission-Essential Task List (METL)

U.S. ARMY

- What does the METL tell you?
 - Tasks on which an organization must train to be proficient in its designed capabilities or assigned mission.
- Why is a METL important?
 - The METL tells you exactly what you must train and practice in order to be proficient as a unit.
- Who determines the METL?
 - The Department of the Army with input from branch proponents (and in some cases the unit)
- Where do you find the METL?
 - <https://atn.army.mil/ATNPortalUI/METL/>



Identifying Key Reference Sources



U.S. ARMY

- FMS Web: <https://fmsweb.fms.army.mil>
- Army site that tells you the Manning and Table of Equipment (MTOE) for units.
 - Tells you what you should have and what you should expect in your formation when it comes to personnel and equipment.
- Can be used to justify getting personnel and equipment.



Identifying Key Reference Sources



U.S. ARMY

- Army Training Network: <https://atn.army.mil>
 - Army site that tells you Mission Essential Task List (METL) and Combined Arms Training Strategies (CATS) for your organization.
 - Tells you what to train and how to train it.
 - Should be used to build your training strategy.



Walking Through an Example



U.S. ARMY

- Where does the EW PLT reside within the average BDE?
 - In HHC BDE Company at ABCT/SBCT and in the Multi-Functional Reconnaissance Company (MFRC) at the MBCT.

Example Overview: A typical MBCT, MFRC METL

1. Go to FMS web (<https://fmsweb.fms.army.mil>) and sign in
2. Click on ORG Box/Line Charts
3. Choose 2-101ABN
4. Identify the UICs for the BDE and MFRC
5. Log into ATN
6. Select the METL button
7. Type in UIC for each example echelons and Identify the METL Tasks



Example

U.S. ARMY

FMSWeb

Quick Search Quick Compare 24.37 Tickets Preferences Connolly.T Guide

No system alerts at this time.

Notices UIC by FY System Inputs System Outputs Doc. Stats Recent Cyber

Date	Tool	Notice
1/8/2026	4610R	Training Notice! Thursday, 8 January 2025 -- 4610R and FMSWeb Briefing 1300-1500 EST. If interested in attending, send an e-mail to the USAFMSA TRAINING COORDINATORS: Mr. Busche [thomas.m.busche.civ@army.mil] / Mr. Humphrey [christopher.d.humphrey8.civ@army.mil] / Mr. Coleman [erik.j.coleman.civ@army.mil].
1/1/2026	4610R	Thursday, 1 January -- New Year's Day
12/25/2025	4610R	Wednesday, 25 December -- Christmas Day
12/23/2025	4610R	12/23/2025 JAN ERVB Board Lock 1200 EST [ANC [AN] / ARNG [GB/NG] / ARSOUTH [6A] / ATEC [AT] / INSCOM [AS] / HQDA [CS/DM/SA/SB/SE/SJ] / USASOC [SP] USMA [MA] and Joint/Defense [DF/DJ/J1/JA]] NOTE: THE NOVEMBER 2025 ERVB CYCLE IS BEING COMBINED WITH THE JANUARY 2026 ERVB CYCLE. The following commands that normally participate in the November ERVB cycle are eligible to participate in the January 2026 cycle: AMC [X1] TRADOC [TA/TC] USARC [AR/HR].

Home
Logout from FMSWeb
Links-References
Training Videos
FMSWeb FAQ
GFIM Notes
UIC Shortcuts

Documentation Tools

- Authorization Docs**
MTOE / TDA / JTA
UIC Hierarchies
Submit 4610R
- Requirements Docs**
TOE / ModPath / BOIP
AMMDB
UTC/TUCHA
- CTA**
8-100 / 50-900
50-909 / 50-970
- Lookup Codes**
Validation codes and definitions.
- TDA 4610R**
TDA Eqp. tracking tool

Analysis Tools

- Reports**
Pos. detail lines
Eqp. detail lines
- Org. Box/Line Charts** (highlighted with a red arrow and box)
UIC Org. charts
SBCT / ABCT / IBCT / more...
- Out of Cycle Report**
Out of Cycle Report

Support Tools

- Text parser
- FMBB Toolset
- Modularity Files
- MTOE/TDA Files/Reports
- FDU/ORDAB Briefings
- FM Contacts
- Public Files



Example

U.S. ARMY

Navigation buttons: CORPS, ARMY CMDs, ASCC, DRU, DIVARTY, ARNG, APS, DIV, MBCT, SBCT, ABCT, IBCT, FA BDE, CAB, ACR, SUST, MI.

Infantry Brigade Combat Teams
Additional Search Options (IBCT):

Please select a structure to view or use the menu options below to refine your search.

Structure: IBCT ▼
FY: 2027 ▼
UIC/Title: 2-101ABD

Search

For all UIC hierarchy questions, please refer to the AOS /GFM DI section of the FMSWeb FAQ pages. This tool is located on the left side of the main tools page.

FY 27 ▼ Hierarchy for Structure (IBCT) w/Structure Derivatives

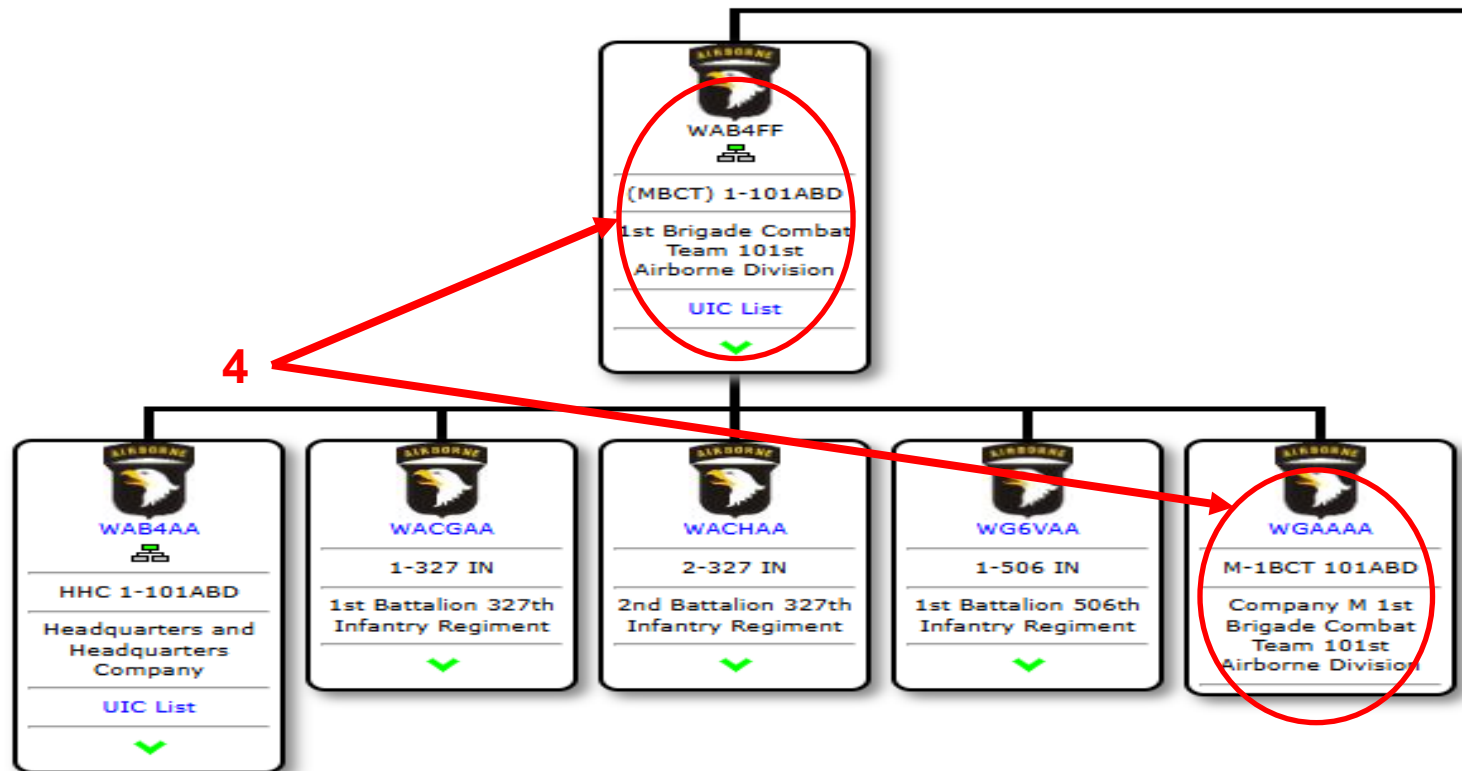
WAB5FF (IBCT) 2-101ABD (2nd Brigade Combat Team)

3



U.S. ARMY

Example





Example



U.S. ARMY

MENU



SEARCH

LOGOUT

Unit Training Management

Home Station Training Enablers

Leader Development

Specific Training Resources

ATIS

CATS

METL

T&EO Task Search

Training Doctrine

OE/OPFOR Resources

Center for the Army Profession and Leadership

Combat Training Centers (CTCs)

UTM Workshop Request

TADSS at Your Local TSC

Command and General Staff College

AFT Guidance and Training Resources

AR 350-1, Appendix C Mandatory Training

Army Space Training Integration - D3SOE

NCO Worldwide

Traumatic Brain Injury (TBI)

More...

SERE Level A Training Support Package, Version 2.0

More...

Counter-Small Unmanned Aircraft Systems (C-sUAS)

More...

More...

Training Doctrine and Publications

Centers of Excellence (CoE) / Proponents

HQDA / Command Training Guidance

Contact

5

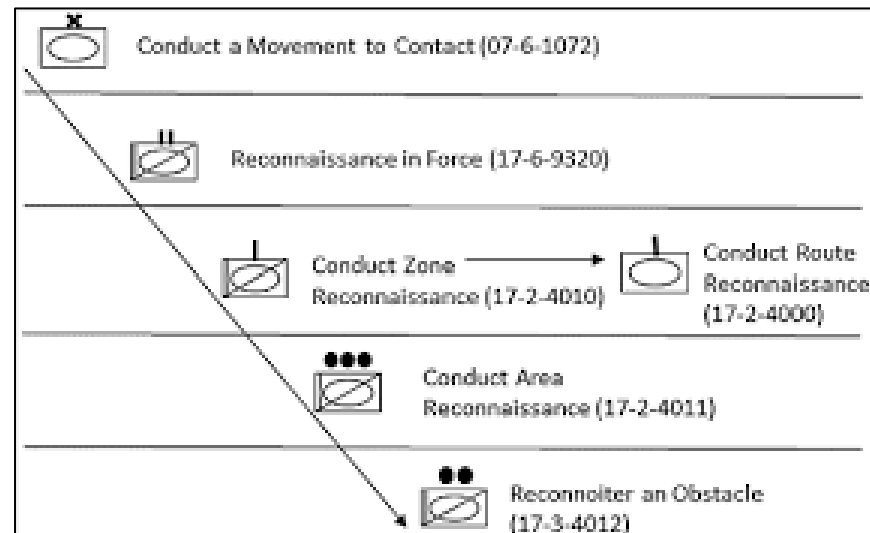
6



METL Nesting

U.S. ARMY

- METLs are different at echelon
 - Due to capabilities and mission sets
- METLs progress from lower to higher echelons
- Each unit has their piece in mission readiness for the parent organization



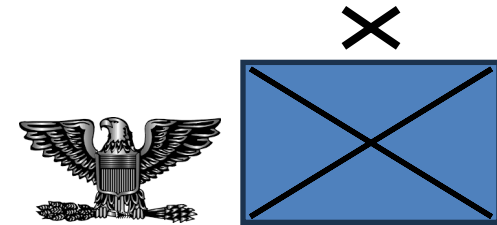


METL Nesting

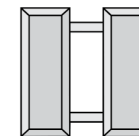


U.S. ARMY

- **Conduct Area Security**
 - **Conduct an Area Defense**
 - **Conduct a Movement to Contact**
 - **Conduct an Attack**
 - **Conduct Expeditionary Deployment Ops**
-
- **Conduct Aerial Reconnaissance Missions**
 - **Conduct Electromagnetic Warfare (EW)**
 - **Manage Information Collection Requirements**
 - **Perform Situation Development**
 - **Conduct Human Intelligence Collection**



MICO METL



HHC BDE
/ MFRC

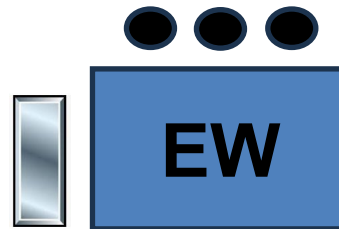


Identify PLT Essential Tasks



U.S. ARMY

- Supported CO METL Task: **Conduct Electromagnetic Warfare (EW)**



- Conduct EW - Supporting Tasks

Step Number	Task Number	Title	Proponent	Status
1.	71-PLT-5100	Conduct Troop Leading Procedures	71 - Mission Command (Collective)	Approved
3.	11-CW-8013	Operate a Combat Network Radio (CNR) System	11 - Signal (Collective)	Approved
3.	07-PLT-5009	Conduct a Rehearsal - Platoon	07 - Infantry (Collective)	Approved
3.	71-PLT-5100	Conduct Troop Leading Procedures	71 - Mission Command (Collective)	Approved
4.	07-PLT-9011	Conduct Tactical Movement in an Urban Area - Platoon	07 - Infantry (Collective)	Approved
4.	07-PLT-1341	Conduct Tactical Movement - Platoon	07 - Infantry (Collective)	Approved
5.	13-PLT-2019	Establish An Electronic Warfare (EW) Site	13 - Cyber (Collective)	Approved
5.	13-TM-2019	Establish an Electromagnetic Warfare (EW) Site	13 - Cyber (Collective)	Approved
5.	05-PLT-3003	Camouflage Equipment	05 - Engineers (Collective)	Approved
6.	13-PLT-2012	Conduct Electronic Attack (EA)	13 - Cyber (Collective)	Approved
7.	13-PLT-2013	Perform Electronic Protection (EP) Actions	13 - Cyber (Collective)	Approved
8.	13-PLT-2014	Provide Electronic Warfare Support (ES)	13 - Cyber (Collective)	Approved
11.	71-PLT-5100	Conduct Troop Leading Procedures	71 - Mission Command (Collective)	Approved



U.S. ARMY



Define Required Individual and Collective Tasks within the EW PLT METL



Understanding Supporting Tasks



U.S. ARMY

- Each Task has a supporting Combined Arms Training Strategy (CATS)
- Within the CATS there is a Training and Evaluation Outline (T&EO)
- Task evaluation

IDV-Marksmanship



TM-React to Contact



SQD-Conduct Attack



PLT-Conduct Attack



Individual

Collective

Foundations

Complexity



Understanding Supporting Tasks



U.S. ARMY

- Supporting tasks:
 - Located at end of T&EO
 - Task inherent to complete the CATS for the identified T&EO task
 - Can be individual or collective
 - Shows what to train to be proficient at the T&EO task



Conduct Electromagnetic Warfare (EW) - Supporting Tasks



U.S. ARMY

Step Number	Task Number	Title	Proponent	Status
1.	71-PLT-5100	Conduct Troop Leading Procedures	71 - Mission Command (Collective)	Approved
3.	11-CW-8013	Operate a Combat Network Radio (CNR) System	11 - Signal (Collective)	Approved
3.	07-PLT-5009	Conduct a Rehearsal - Platoon	07 - Infantry (Collective)	Approved
3.	71-PLT-5100	Conduct Troop Leading Procedures	71 - Mission Command (Collective)	Approved
4.	07-PLT-9011	Conduct Tactical Movement in an Urban Area - Platoon	07 - Infantry (Collective)	Approved
4.	07-PLT-1341	Conduct Tactical Movement - Platoon	07 - Infantry (Collective)	Approved
5.	13-PLT-2019	Establish An Electronic Warfare (EW) Site	13 - Cyber (Collective)	Approved
5.	13-TM-2019	Establish an Electromagnetic Warfare (EW) Site	13 - Cyber (Collective)	Approved
5.	05-PLT-3003	Camouflage Equipment	05 - Engineers (Collective)	Approved
6.	13-PLT-2012	Conduct Electronic Attack (EA)	13 - Cyber (Collective)	Approved
7.	13-PLT-2013	Perform Electronic Protection (EP) Actions	13 - Cyber (Collective)	Approved
8.	13-PLT-2014	Provide Electronic Warfare Support (ES)	13 - Cyber (Collective)	Approved
11.	71-PLT-5100	Conduct Troop Leading Procedures	71 - Mission Command (Collective)	Approved

OPFOR Task(s): None

Supporting Individual Task(s): None

Supporting Drill(s):

Step Number	Drill Number	Drill Title	Drill Type	Proponent	Status
4.	07-SQD-D9501	React to Direct Fire Contact While Dismounted - Squad	Battle Drill	07 - Infantry (Collective)	Approved
4.	07-SQD-D9504	React to Indirect Fire While Dismounted - Squad	Battle Drill	07 - Infantry (Collective)	Approved
4.	07-PLT-D9508	Establish Security at the Halt - Platoon	Battle Drill	07 - Infantry (Collective)	Approved
4.	07-SQD-D9505	Break Contact - Squad	Battle Drill	07 - Infantry (Collective)	Approved
4.	07-SQD-D9502	React to Ambush (Dismounted) - Squad	Battle Drill	07 - Infantry (Collective)	Approved



Practical Exercise (Identify EW PLT Tasks)



U.S. ARMY

- Download the T&EO for Conduct EW/CYBER from ATN
- Pick 2-3 supporting task identified with in the T&EO
- Look up the T&EO for each of the supporting task you identified and find the supporting task for each
- Continue to look up the T&EO for each new task you identify and keep a list of each supporting collective and individual task you find
- Ask yourself the following questions when you identify a task:
 - Are these tactical tasks or technical tasks?
 - Are these related to EW/Cyber or are they general army soldier/leader tasks?
 - Which task occur multiple time and why do you think that?

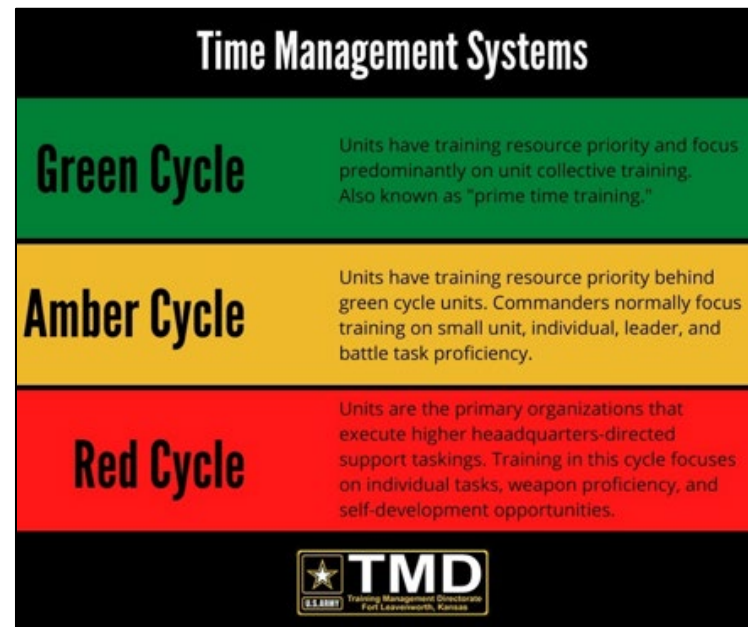


What if your EW PLT does not have a METL?



U.S. ARMY

- The platoon leader and platoon sergeant conduct a mission-essential task list crosswalk (FM 7-0).
- Training plan and task list must align to support the higher echelons' training plan and long-range training calendar.



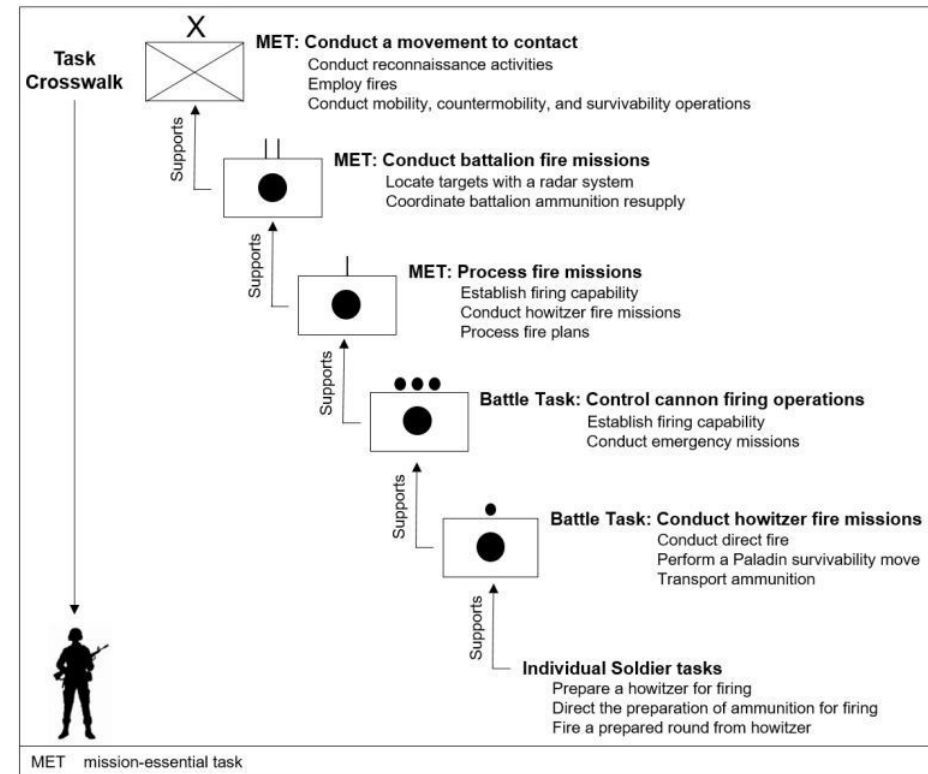


Conducting a Cross Walk



U.S. ARMY

- Conducting a cross walk requires:
 - A supported unit Mission
 - A clear CDR's intent
 - Understanding of the unit MTOE





Conducting a Cross Walk



U.S. ARMY

- Do's and Don'ts
 - DO:
 - Have an open dialogue with subordinate and superior leaders
 - Research the problem before bringing it to you CDR
 - Have a proposed solution before elevating it to higher echelons
 - Have a flushed out METL before starting a training plan
 - Don't
 - Let higher dictate without input from the EW PLT
 - Elevate problems without a proposed way ahead
 - Start training on your assumptions only
 - Wait for some one else to solve the problem



Example Cross Walk



U.S. ARMY

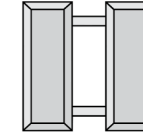


ID Priority METL Task

Conduct an Attack

CDRs intent

"I want our unit to be aggressive and focus on the offensive"



ID Priority METL Tasks

Manage Information Collection Requirements

Perform Situation Development

CDRs intent

"I want PLTs to focus on supporting maneuver by building situational awareness through ISR"



INPUTs:

- BDE wants to Focus on offense
- MFRC wants to focus on ISR
- I have limited jamming equipment
- I have good Direction-Finding equipment

Recommended PLT TASK:

Conduct Electromagnetic Support

Training Plan:

Supporting collective and Individual tasks within Conduct Electromagnetic Support

Cyber and Electromagnetic Warfare Training Strategy (CEWTS) [TC 3-12.1.2.2]



U.S. ARMY

- Provides training standards for EW forces with capabilities
- Supports the 2018 National Defense Strategy guidance
 - “Gain and maintain information advantage, particularly in cyberspace, space, and the electromagnetic spectrum.”
- Includes procedures for establishing, implementing, and assessing operations with training requirement standards for operational performance





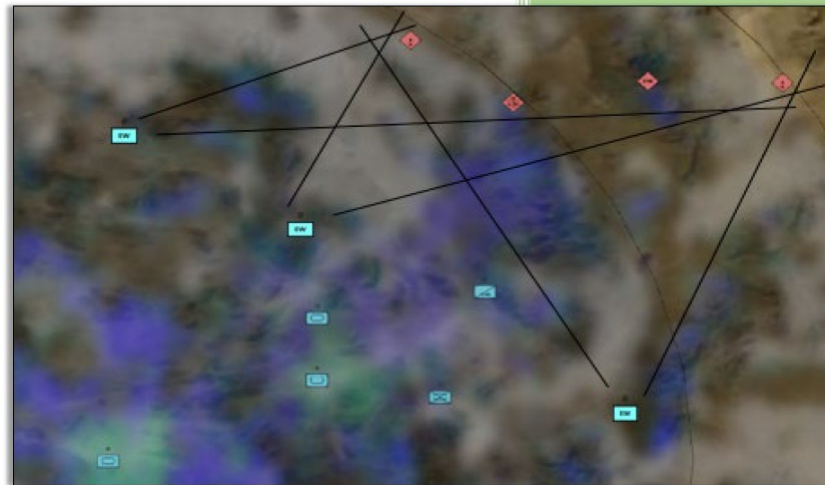
FORSCOM's Electromagnetic Warfare Training Strategy (EWTS)



U.S. ARMY

- Establishes EW training and certification guidance for FORSCOM units that have received quick reaction capabilities (QRC).
- Adaptive to the emerging EW capabilities needed to enhance lethal and non-lethal operations during near-peer or large-scale combat operations (LSCO).

Electromagnetic Warfare Training Strategy (EWTS)





When does the EW Tech get involved?



U.S. ARMY

- Overall responsibility of training is upon the EW PLT leadership.
- Assisting with resourcing training items that the EW PLT can not acquire organically.
- Assisting with coordination between BDE and BN training events that request EW support, which is then published in the BDE Order/FRAGO.
- CEWTS training validation.





CEWTS Validator Responsibilities



U.S. ARMY

- Identify the number of evaluators needed for each element.
- Ensure evaluators are properly trained and standardized through an Evaluator Academy.
- Manage supporting documents for certification and configuration standards.
- Conduct spot checks during certification.
- Plan reset requirements in conjunction with the certifier during Tier rehearsals.
- Develop certification scenarios based on current or previous CTC rotation scenarios or BDE FTX scenarios.

Equipment and Support Considerations – EW PLT MTOE



U.S. ARMY

- How many Soldiers are in the EW PLT?
- What equipment does the EW PLT have?
- Example Walk Through:
 1. Go to FMS web <https://fmsweb.fms.army.mil>
 2. Enter the UIC for MFRC you had earlier
 3. Go to sub teams and find the EW PLT
 4. Explore the EW PLT MTOE

MBCT, MFRC, EW PLT Manning Example



U.S. ARMY

110 - ELECTROMAGNETIC WARFARE

P A R N O					G R A D E	S Q I D	A A I I	A S S I C N	L L P R D P S P	I P P P P P	REQ STR	AUTH STR	PAR UNIT REQ STR	PAR UNIT AUTH STR	R M K 1	R M K 2	R M K 3	C R D M U B S C C O	D S A M B S C C O	M D E P	SDTC	CAF	MM	
110	01	PLATOON LEADER	O2	17B00						CY	K	A	E	7						11101400	W501	AAT	M415	A
110	02	PLATOON SERGEANT	E7	17E40						CY	I	A	E	7						11101400	W501	AAU	M415	A
110	03	EW SPEC	E4	17E10						CY	I	A	E	7						11101400	W501	DQH	M415	A
Para. Count												3	3	3	3									

111 - ELECTROMAGNETIC WARFARE

P A R N O			G R A D E	POS	S Q I D	A S S I C I C N C	L P R D P S P	L I N C H	B R N C H	I D N S T T	P P S S R R	P P S S R R	REQ STR	AUTH STR	PAR UNIT REQ STR	PAR UNIT AUTH STR	R M K 1	R M K 2	R M K 3	C R D M U B S C C O	D S U B S C C O	A M B S C C O	M D E P	SDTC	CAF	MM
111	01	SR TEAM CHIEF	E6	17E30					CY	I	A	E	7	3	3	3	3	BB			11101400	W501	APT	M415	A	
111	02	EW SERGEANT	E5	17E20					CY	I	A	E	7	3	3	3	3				11101400	W501	MRS	M415	A	
111	03	EW SPEC	E4	17E10					CY	I	A	E	7	3	3	3	01	15			11101400	W501	DQH	M415	A	
Para. Count												9	9	9	9											

112 - ELECTROMAGNETIC WARFARE

P A R N O				G R A D E	S Q I D	A A I I	A S S I C N	L P R D	L B R N C H	I D E N T I T	P P S R R Q	P P S R R Q	REQ STR	AUTH STR	PAR UNIT REQ STR	PAR UNIT AUTH STR	R M K 1	R M K 2	R M K 3	C R D M U B S C O	D S U M B S C O	M D E P	SDTC	CAF	MM		
112	01	TEAM CHIEF		E5	17E20					CY	I	A	E	7	6	6	6	6	BB			11101400	W501	ABV	M415	A	
112	02	EW SPEC		E4	17E10					CY	I	A	E	7	6	6	6					11101400	W501	DQH	M415	A	
112	03	EW SPEC		E3	17E10					CY	I	A	E	7	6	6	6	01	15			11101400	W501	DQH	M415	A	
Para. Count															18	18	18	18									
															139	139	139	139									

MBCT, MFRC, EW PLT Equipment Example



U.S. ARMY

110 - ELECTROMAGNETIC WARFARE										
P A R T N O	LIN	E R C	NOMENCLATURE	REQ EQ	AUTH EQ	PARENT UNIT REQ EQ	PARENT UNIT AUTH EQ	R M K 1	C R D M U K I 2 C	S U B C O
110	B49004	A	BAYONET MULTIPURPOSE SYSTEM: XM9	3	3	3	3		IBM001	
110	B57077	A	BATTLEFIELD ANTI-INTRUSION SYSTE: AN/PRS-9	1	1	1	1		IAM103	
110	B67766	A	BINOCULAR: MODULAR CONSTRUCTION MIL SCALE RETICLE 7X50MM W/E	1	1	1	1		LAM021	
110	C05002	A	COMPUTER SYS DIGITAL: AN/PYQ-10(C)	1	1	1	1		CAM304	
110	C05036	A	COMPUTER SET DIGITAL (JBC-P): AN/UYSK-128B(V)3	1	1	1	1		FAM006	
110	C06935	A	CARBINE 5.56 MILLIMETER: M4A1	3	3	3	3		IAM151	
110	E05008	A	ENCRYPTION-DECRYPTION EQUIP: KGV-72	1	1	1	1		FAM024	
110	E05037	A	Electronic Warfare Planning and Management Tool	1	1	1	1		CAM442	
110	G18358	A	GEN SET: DED SKID MTD 3KW 60HZ	1	1	1	1 TEJ		OAM926	
110	J00697	A	JOINT CHMCL AGENT: DETECTOR	1	1	1	1		GAM024	
110	M05023	A	MULTIMETER AN/GSM-437: ()	1	1	1	1		PAM228	
110	M05098	A	Mounted Mission Command- Software (MMC-S) Application	1	1	1	1		FAM049	
110	M12986	A	MASK CHEMICAL BIOLOGICAL JOINT SERVICE GENERAL PURPOSE: FIELD M50	3	3	3	3		GAM002	
110	M79678	A	MONOCULAR NIGHT VISION DEVICE: AN/PVS-14	3	3	3	3		IAM058	
110	N96248	A	NAVIGATION SET: SATELLITE SIGNALS AN/PSN-13	1	1	1	1		CAM461	
110	R31061	A	RADIAC SET: AN/UDR-13	1	1	1	1		GAM021	
110	R44999	A	RADIO SET: AN/VRC-89F(C)	1	1	1	1		CAM006	
110	R45543	A	RADIO SET: AN/VRC-92F(C)	2	2	2	2		CAM525	
110	R87139	A	RADIO SET: AN/VRC-104(V)6 150 WATT W/ PRC-150 HF RADIO	1	1	1	1		CAM556	
110	S60288	A	SIGHT: REFLEX COLLIMATOR	3	3	3	3		IAM064	
110	T56383	A	TRUCK UTILITY EXPANDED CAPACITY ENHANCED 4X4: M1165A1	1	1	1	1 TEP		TAM220	
110	W37483	A	TOOL KIT ELECTRIC EQUIPMENT: TK-101/GSQ	1	1	1	1		OAM443	
110	Z05987	A	RADIO SET: AN/PRC-158A(V)4	1	1	1	1		CAM539	
111 - ELECTROMAGNETIC WARFARE										
P A R T N O	LIN	E R C	NOMENCLATURE	REQ EQ	AUTH EQ	PARENT UNIT REQ EQ	PARENT UNIT AUTH EQ	R M K 1	C R D M U K I 2 C	S U B C O
111	B49004	A	BAYONET MULTIPURPOSE SYSTEM: XM9	9	9	9	9		IBM001	
111	C05036	A	COMPUTER SET DIGITAL (JBC-P): AN/UYSK-128B(V)3	3	3	3	3		FAM006	
111	C06935	A	CARBINE 5.56 MILLIMETER: M4A1	6	6	6	6		IAM151	
111	E05008	A	ENCRYPTION-DECRYPTION EQUIP: KGV-72	3	3	3	3		FAM024	
111	E05037	A	Electronic Warfare Planning and Management Tool	3	3	3	3		CAM442	
111	L69080	A	LAUNCHER GRENADE: M320A1	3	3	3	3		IAM020	
111	M05098	A	Mounted Mission Command- Software (MMC-S) Application	3	3	3	3		FAM049	
111	M09009	A	MACHINE GUN 5.56 MILLIMETER: M249	3	3	3	3		IAM002	
111	M12986	A	MASK CHEMICAL BIOLOGICAL JOINT SERVICE GENERAL PURPOSE: FIELD M50	9	9	9	9		GAM002	
111	M79678	A	MONOCULAR NIGHT VISION DEVICE: AN/PVS-14	9	9	9	9		IAM058	
111	N96248	A	NAVIGATION SET: SATELLITE SIGNALS AN/PSN-13	3	3	3	3		CAM461	
111	R31061	A	RADIAC SET: AN/UDR-13	3	3	3	3		GAM021	
111	R45543	A	RADIO SET: AN/VRC-92F(C)	3	3	3	3		CAM024	
111	R45543	A	RADIO SET: AN/VRC-92F(C)	9	9	9	9		CAM525	
111	R87139	A	RADIO SET: AN/VRC-104(V)6 150 WATT W/ PRC-150 HF RADIO	3	3	3	3		CAM556	
111	S45729	A	SIGHT BORE OPTICAL: M150	3	3	3	3		IAM116	
111	S60288	A	SIGHT: REFLEX COLLIMATOR	6	6	6	6		IAM064	
111	S90535	A	MEDIUM WEAPON THERMAL SIGHT (MWTS): AN/PAS-13(V)2	3	3	3	3		IAM055	
111	T56383	A	TRUCK UTILITY EXPANDED CAPACITY ENHANCED 4X4: M1165A1	3	3	3	3 TEP		TAM220	
111	Z05987	A	RADIO SET: AN/PRC-158A(V)4	3	3	3	3		CAM539	



MBCT, MFRC, EW PLT Equipment Example



U.S. ARMY

112 - ELECTROMAGNETIC WARFARE										
P A R N O	LIN	E R C	NOMENCLATURE	REQ EQ	AUTH EQ	PARENT UNIT REQ EQ	PARENT UNIT AUTH EQ	R M K 1	C R D M U K I 2 C	S U B C O
112	B49004	A	BAYONET MULTIPURPOSE SYSTEM: XM9	18	18	18	18		IBM001	
112	C05036	A	COMPUTER SET DIGITAL (JBC-P): AN/UYK-128B(V)3	6	6	6	6		FAM006	
112	C06935	A	CARBINE 5.56 MILLIMETER: M4A1	12	12	12	12		IAM151	
112	E05008	A	ENCRYPTION-DECRYPTION EQUIP: KGV-72	6	6	6	6		FAM024	
112	E05037	A	Electronic Warfare Planning and Management Tool	6	6	6	6		CAM442	
112	L69080	A	LAUNCHER GRENADE: M320A1	6	6	6	6		IAM020	
112	M05098	A	Mounted Mission Command- Software (MMC-S) Application	6	6	6	6		FAM049	
112	M09009	A	MACHINE GUN 5.56 MILLIMETER: M249	6	6	6	6		IAM002	
112	M12986	A	MASK CHEMICAL BIOLOGICAL JOINT SERVICE GENERAL PURPOSE: FIELD M50	18	18	18	18		GAM002	
112	M79678	A	MONOCULAR NIGHT VISION DEVICE: AN/PVS-14	18	18	18	18		IAM058	
112	N96248	A	NAVIGATION SET: SATELLITE SIGNALS AN/PSN-13	6	6	6	6		CAM461	
112	R31061	A	RADIAC SET: AN/UDR-13	6	6	6	6		GAM021	
112	R45543	A	RADIO SET: AN/VRC-92F(C)	6	6	6	6		CAM024	
112	R45543	A	RADIO SET: AN/VRC-92F(C)	18	18	18	18		CAM525	
112	R87139	A	RADIO SET: AN/VRC-104(V)6 150 WATT W/ PRC-150 HF RADIO	6	6	6	6		CAM556	
112	S45729	A	SIGHT BORE OPTICAL: M150	6	6	6	6		IAM116	
112	S60288	A	SIGHT: REFLEX COLLIMATOR	12	12	12	12		IAM064	
112	S90535	A	MEDIUM WEAPON THERMAL SIGHT (MWTS): AN/PAS-13(V)2	6	6	6	6		IAM055	
112	T56383	A	TRUCK UTILITY EXPANDED CAPACITY ENHANCED 4X4: M1165A1	6	6	6	6 TEP		TAM220	
112	Z05987	A	RADIO SET: AN/PRC-158A(V)4	6	6	6	6		CAM539	



EW Platoon Equipment

U.S. ARMY

Weapons:

M4
M249
M320

Vehicles:

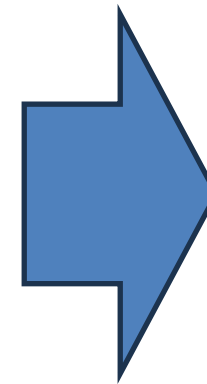
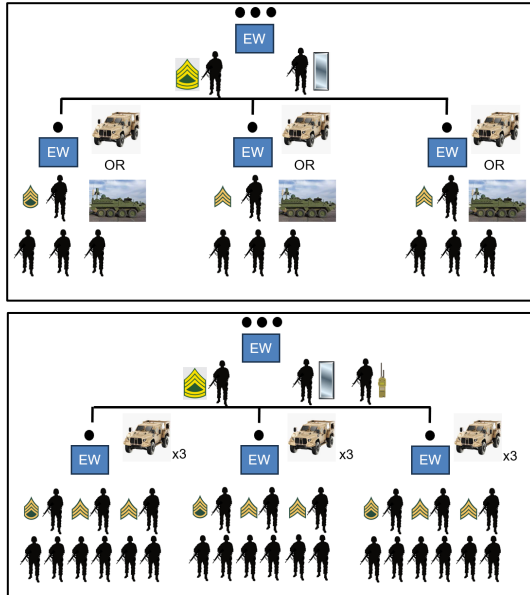
Stryker
HMMWV/JLTV

Communications:

ASIP
HF
TACSAT
JBCP

EW Equipment:

EWPMT
COTS (VROD/BEAST)



How Does this
impact a
training plan?



Understanding your EW Equipment



U.S. ARMY

- Army EW Program of Record Equipment:
 - Electronic Warfare Planning and Management Tool (EWPMPT) v1
- Most EW PLTs will have some combination of COTS or GOTS equipment.
- Each equipment type will have different capabilities and limitations.





EW Platoon Equipment

U.S. ARMY

- Conduct an ES mission:
 - How do you get there?
 - Mounted = Driver's training
 - Dismounted = Land Navigation training
 - How do you establish security?
 - Collective = Conduct security at the halt
 - Individual = Qualify on weapon systems
 - How do you communicate ES information?
 - Collective = Conduct TLPs
 - Individual = Operate assigned radio system



EW Platoon Training



U.S. ARMY

- Train more than just EW tasks
- Shoot, Move, and Communicate





U.S. ARMY



Apply the 8-step training model to EW PLT METL



Review



U.S. ARMY

- What is the 8 step training model?
 - The Army's preeminent training model.
- What are the steps?





Step 1 – Plan the Training Event



U.S. ARMY

- Develop specific, attainable, and measurable training objectives
 - Commander's training guidance determined from METL Cross Walk
 - Evaluation criteria determined by T&EO from CATS.
- Allocate and schedule enough time for the event
- Create scenarios and instructions to support the evaluated training objectives
- Identify support and resources needed
- Identify potential training hazards and mitigate associated risks



Step 1 – Plan the Training Event



U.S. ARMY

- Identify frequencies required and coordination needed
 - Frequency allocation for target emitters
 - OCONUS/CONUS EA request for EW equipment





Step 2 – Train and Certify Leaders



U.S. ARMY

- Train and certify leaders
 - Identify the supporting Individual tasks
- Certified leaders train and certify other leaders
- Qualified personnel demonstrate performing the training task(s) to the standard
- Train-the-trainer concept ensures leaders and trainers can effectively instruct and certify the unit





Step 3 - Recon Training Sites



U.S. ARMY

- Leaders conduct a reconnaissance of proposed training areas and facilities.
- Leader's reconnaissance verifies the location can support the proposed training and allow the unit to achieve its training objectives.
- Verify coordination, scheduling, and preparation of resources, training areas, and training support plans
 - Range Facility Management Support System (RFMSS) for TA/Range availability
- Not properly planned, coordinated, supported, and resourced training events are unlikely to meet the established training objectives.



Step 4 – Issue the Operation Order (OPORD)



U.S. ARMY

- OPORD
 - Ensures subordinates have the information needed to execute the training event.
 - Clearly identifies tasks to be trained and training objectives
 - Clearly understandable mission statement
 - Communicates the scope of the training, how to conduct the event, and tasks to train and evaluate
- Conduct a back brief to ensure your subordinate leaders understand the mission.





Step 5 – Rehearse

U.S. ARMY

- Rehearsals are critical to the execution of any operation or training plan
- All personnel involved in the training event participate in rehearsals
- Company and platoon level leaders supervise training rehearsals





Step 6 - Train

U.S. ARMY

- Maximum participation
 - Units can not be certified without appropriate leader participation
 - Leader requirements are within the CATS T&EO
- Minimizing training distractors
 - Do not plan training after or right before holidays
 - Ensure you have deconflicted any competing requirements





Step 6 - Train

U.S. ARMY

- Leader involvement
 - Check and supervise training
 - T&EO's on site if possible
 - Spot check trainers
 - Sit through a task and ask yourself, "Is this what we wanted?"



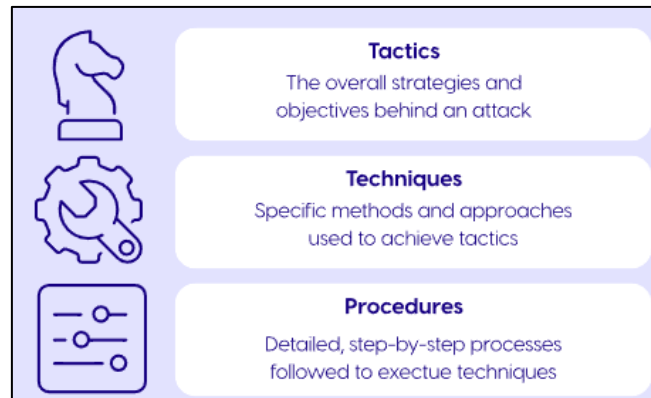


Step 7 – Conduct After Action Reviews



U.S. ARMY

- During and after training:
 - Review the tasks trained
 - Assess the unit's training level compared to the training objective
 - Document local lessons learned
 - Refine tactics, techniques, and procedures
 - SOP development is key for lessons learned
 - SOPs help units retain knowledge during leader transitions





Step 7 – Conduct After Action Reviews



U.S. ARMY

- Record assessments in the Army Training Information System (ATIS) Website (<https://learn.atis.army.mil/>)
- For use in future training events and inclusion in the unit SOP
- Record results to help identify short falls or other patterns
- Records can help with future developmental counseling, evaluations, and awards





Step 8 – Retrain

U.S. ARMY

- Units often neglect retraining:
 - Time
 - Resource limitations
 - Conflicting scheduling requirements
- Train to established standards
- Leaders must honestly and objectively assess their unit's ability to meet performance standards





U.S. ARMY



Apply Troop Leading Procedures (TLP) to the EW PLT



Troop Leading Procedures (TLPs)



U.S. ARMY

- Provide a framework for planning and preparing for missions
- Smaller units use TLPs to plan and prepare for operations
- Platoon level, the platoon leader assumes the responsibility for planning to solve tactical problems
 - Platoon sergeant and team chiefs support the PL
 - Type, amount, and timeliness of information passed from higher to lower directly affects the platoon leader's TLPs



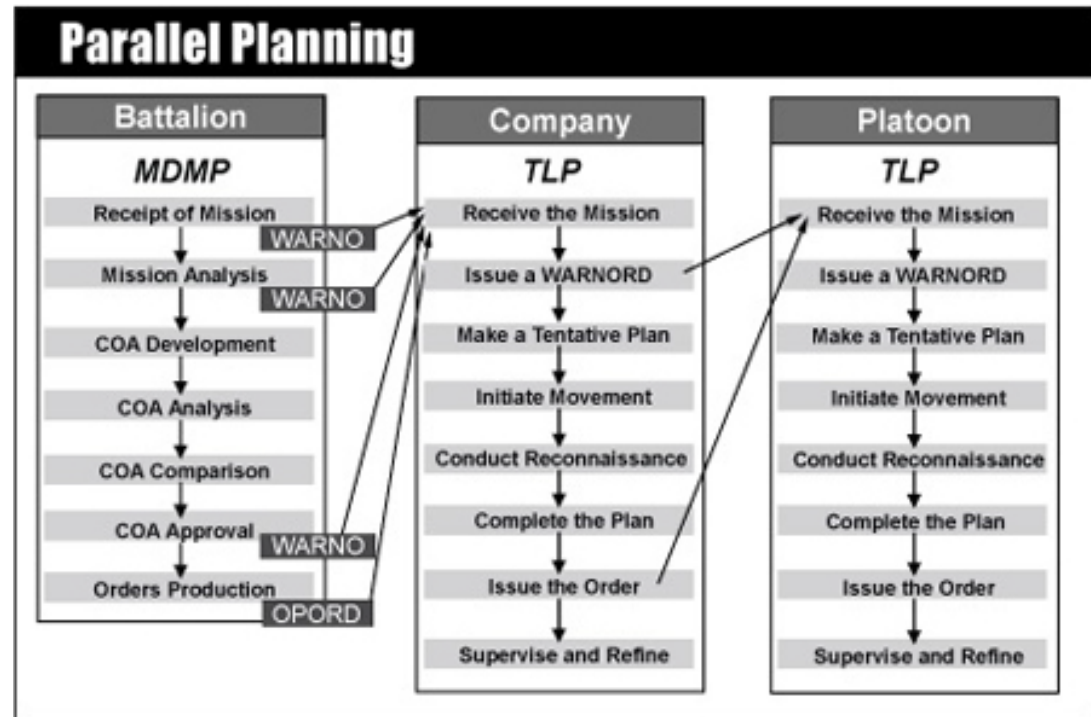


Eight Steps of TLPs



U.S. ARMY

1. Receive the Mission
2. Issue a WARNORD
3. Make a tentative plan
4. Initiate movement
5. Conduct reconnaissance
6. Complete the plan
7. Issue the order
8. Supervise and refine





Step 1 – Receive the Mission



U.S. ARMY

- Receive the mission through a written or verbal
 - WARNORD
 - OPORD
 - FRAGORD
- Determine the time available
 - Plan
 - Prepare
 - Execute
- Determine the one-third–two-thirds timeline for leader planning and subordinate preparation
- Prepare an initial planning timeline





Step 2 – Issue the Warning Order



U.S. ARMY

- WARNORD include the following items:
 - Current threat situation
 - Higher headquarters mission or concept of the operation
 - Commander's intent
 - Initial operational timeline
 - Platoon mission
 - Updated graphics
 - Reconnaissance to initiate
 - Movement to initiate
 - Earliest time of movement

SITUATION

- Area of Interest
- Area of Operations
- Enemy Forces
- Friendly Forces
- Attachments and Detachments

MISSION

- Who
- What
- When
- Where
- Why

EXECUTION

- Concept of Operations
- Tasks to Subordinate Units
- Coordinating Instructions

SUSTAINMENT

- Logistics
- Personnel Services Support
- Army Health Systems Support

COMMAND AND SIGNAL

- Command
- Control
- Signal



Step 2 – Issue the Warning Order



U.S. ARMY

- Planning and preparation instructions
- Commander's critical information requirements
- Changes to task organization
- Use of specialized equipment
- Attachment of enablers
- Recommended supply load
- Rehearsal of key events
- Readiness condition and vehicle preparation schedule
- Personal protective equipment modifications
- Time and place for issuing the OPORD





Step 3 – Make a Tentative Plan



U.S. ARMY

- Mission Analysis
 - Helps the leadership determine:
 - Current situation
 - Platoon's mission
 - How to best accomplish the mission
 - Possible risks
 - Mission-based spectrum observation requirements for ES
 - Calculated effective received power at target for EA





Step 3 – Make a Tentative Plan



U.S. ARMY

- Conduct mission analysis by evaluating the mission variables of METT-TC (I)
- Mission Variables
 - **M**ission
 - **E**nemy
 - **T**errain
 - **T**roops and support available
 - **T**ime available
 - **C**ivil considerations
 - **I**nformational considerations





Step 4 – Initiate Movement

U.S. ARMY

- Leadership initiate movements
 - Continue mission preparation
 - Posture the unit for starting the mission
- Take place any time during the sequence of troop leading procedures
- Includes:
 - Execution of priorities of work
 - Movement to an assembly area, battle position, or a new AO
 - Movement of guides or quartering parties



Step 5 - Conduct Reconnaissance



U.S. ARMY

- EW site reconnaissance involves:
 - Reps from the parent unit G-3/S-3
 - Key EW leaders
- All key leaders from the platoon participate in site reconnaissance and maintain radio communications
- Proposed site should be large enough to accommodate all EW elements with their support equipment





Step 6 – Complete the Plan



U.S. ARMY

- Refine the plan, prepare analog and digital overlays, complete sustainment requirements, and update the tentative plan based on the latest reconnaissance
- Five-paragraph OPORD format helps explain all aspects of the operation:
 - **Situation**
 - Terrain
 - Enemy
 - Higher and adjacent friendly units
 - **Mission**



Step 6 – Complete the Plan



U.S. ARMY

- Execution
- Team tasks
- Team purposes
- Team priorities
- Sustainment
- Command and signal

1. SITUATION

- Area of Interest
- Area of Operations
 - Terrain
 - Weather
- Enemy Forces
 - Latest Threat Intelligence
- Friendly Forces
 - Two Levels Up
 - One Level Up
 - Adjacent Units
- Attachments and Detachments
 - Who/Why

2. MISSION

- Who
- What
- When
- Where
- Why

3. EXECUTION

- Commander's Intent
- Concept of Operations
- Scheme of Movement and Maneuver
 - Describe from Start to Finish
 - Fires
 - CASEVAC
- Tasks to Subordinate Units
- Coordinating Instructions
 - Time Schedule
 - CCIR (PIR, FFIR), EEFI
 - Risk Reduction Control Measures
 - ROE
 - Environmental Considerations
 - Force Protection

4. SUSTAINMENT

- Logistics
 - Maintenance
 - Transportation
 - Field Services
- Personnel Services Support
 - Handling of EPW
- Army Health System Support
 - MEDEVAC
 - Preventive Medicine

5. COMMAND AND SIGNAL

- Command
 - Location of Leaders
- Control
 - Command Post Location
- Signal
 - Radio Frequencies
 - Passwords/Running Pass-words
 - Pyrotechnic Signals

Legend:

CASEVAC= Casualty Evacuation

CCIR= Commander's Critical Information Requirements

EEFI= Essential Elements of Friendly Information

EPW= Enemy Prisoner Of War

FFIR= Friendly Force Intelligence Requirements

PIR= Priority Intelligence Requirements

MEDEVAC= Medical Evacuation

ROE= Rules Of Engagement



Step 7 – Issue the Order

U.S. ARMY

- OPORD explains the leadership's concept of how the unit will accomplish its mission
- Leadership use a terrain model, sand table, sketches, or a map to orient the platoon
- Whenever possible, platoon leadership issue the order in person
- Platoon leadership complete the order with a confirmation brief
 - Team chiefs should be able to backbrief
 - Platoon's mission and intent
 - The commander's intent
 - Their own tasks and purposes
 - When they will inform their teams of the mission details



Step 8 – Supervise and Refine



U.S. ARMY

- After issuing the OPORD, the platoon leadership ensure Soldiers complete the required activities and tasks before mission execution
- Supervision is the primary responsibility of all leadership
- Platoon leadership must check those items or events deemed important for mission accomplishment
 - Conduct backbriefs on all aspects of the platoon mission.
 - Ensure the second-in-charge in each element is prepared to execute in the leader's absence
 - Observe rehearsals



Step 8 – Supervise and Refine



U.S. ARMY

- Inspect load plans to ensure teams carry what is necessary for the mission or what the OPORD specifies
- Inspect the status and serviceability of weapons and communications systems
- Inspect maintenance activities
- Ensure local security is maintained

LAST NAME, FIRST NAME, MI:	
INDIVIDUAL PCC - GARRISON ☐ HAIRCUT ☐ SHAVE (IF APPLICABLE) ☐ CAP, PATROL ☐ COAT AND TROUSERS, ACU ☐ T-SHIRT, SHORT SLEEVE, TAN 499 ☐ UNDERGARMENTS ☐ BELT, RIGGER, TAN 499 ☐ SOCKS, BOOT (TAN, GREEN, OR BLACK) ☐ BOOTS, COMBAT, COYOTE (HOT WEATHER <OR> TEMPERATE) ☐ (2) IDENTIFICATION TAGS WITH CHAIN ☐ CAC (COMMON ACCESS CARD) CHECK EXPIRATION DATE ☐ DRIVER'S LICENSE, MILITARY ☐ PEN / PENCIL AND ☐ NOTEPAD ☐ COPY OF DRILL LETTER  VERSION 1.53 260103	INDIVIDUAL PCC - FIELD ☐ HELMET, ADVANCED COMBAT (ACH) W / LEVER BRACKET ☐ EYE PROTECTION W / CLEAR AND DARK LENS ☐ EAR PROTECTION ☐ GLOVES, COMBAT ☐ SPCS (SOLDIER PLATE CARRIER SYSTEM) ☐ TAP (TACTICAL ASSAULT PANEL) ☐ 5.56 MM UBL: 210 RDS / (7) MAGAZINES ☐ IFAK (INDIVIDUAL FIRST AID KIT) ☐ (2) TOURNIQUETS ☐ WEAPONS CLEANING KIT ☐ FLASHLIGHT W / RED LENS AND BATTERIES ☐ HYDRATION SYSTEM, 3 LITERS, FULL <OR> ☐ (2) 1 LITER CANTEENS, FULL ☐ M50 MASK, CHEMICAL-BIOLOGICAL ☐ M4 CARBINE, 5.56 MM W / SLING ☐ WEAPON OPTIC ☐ NVD <OR> NVG W / VIEWER BRACKET AND ADAPTER TIE-DOWN



Summary

U.S. ARMY

- Identified Mission Essential Task List (METL) of the EW PLT
- Defined Required Individual and Collective Tasks within the EW PLT METL
- Applied the 8-step training model to EW PLT METL
- Applied Troop Leading Procedures (TLP) to the EW PLT



U.S. ARMY



Questions